AI-Driven Design Optimization for Individuals with Spinal Cord Injury

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Abstract— **People with cervical spinal cord injury at C5–C7 retain partial wrist extension but lose finger function. Passive dorsal graspers exploit this residual motion to restore grasp, yet most prescribed devices are abandoned within months by its user because geometry that works for one user fails for another. Wrist extension range and available torque vary by nearly 3:1 across the same injury level, making one-size-fits-all design inadequate and manual clinical fitting too slow to scale.**

We built a three-phase, LLM-driven pipeline that automatically finds personalized dorsal grasper configurations for individual users. Starting from a 15-parameter design space covering finger chain geometry, spread angle, and mounting position, the system iterates between MuJoCo physics simulation and a language model. The LLM interprets structured failure modes from simulation and proposes targeted geometry changes constrained to each user's measured joint limits and force ceiling. We ran experiments across four participant profiles (H01–H04), testing robustness to initialization, LLM nondeterminism, and iteration budget. The pipeline converged to a universal design, passing all eight grasping tasks within each user's force limit in 83% of runs, with the fastest convergence at two Phase 3 iterations. Random parameter search produced zero universal designs in 5000 attempts over the same search space.

The LLM's geometry-aware reasoning is genuinely effective; it compensates for poor initialization and diagnoses failure modes that scalar optimization cannot. But it also anchors: the dominant failure mode across all non-converging runs is the LLM fixing on a suboptimal finger spread angle and failing to escape over 100 iterations. Separately, noise robustness testing shows that Phase 2 cross-validation designs are far more noise-tolerant than Phase 3-optimized designs, because Phase 3 inadvertently drives geometry to the force limit, eliminating noise margin.

Future work should address both findings: simulation-to-real implementation, prompt injection to prevent parameter anchoring, a noise-aware success criterion for Phase 3, and physical fabrication to validate simulation results against real-world force measurements.

I. Introducing Large Language Models into the Assistive Device Space

Assistive wearable devices provide a non-invasive solution for individuals with impaired motor or sensory function, particularly those with preserved upper-limb structure but limited motor control, such as individuals with cervical spinal cord injury (SCI). A majority of patients with injuries at this level cite regaining arm and hand function as their main priority [1]. These devices can substantially enhance functional capabilities, including grasping and object manipulation [2], [3]. A wide range of upper-limb assistive devices has therefore been proposed, which can be broadly characterized along two axes: the level of actuation and the degree of task generalization.

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Along the actuation axis, devices are categorized as passive or active. Passive systems rely on mechanical linkages to augment residual motor function—for example, wrist-driven orthoses that amplify wrist extension to enhance tenodesis grasp in individuals with SCI [4], [5]. Active systems, in contrast, incorporate actuators to drive finger motion directly [6] or introduce supernumerary robotic fingers to supplement grasping capability [7], [8]. Along the task generalization axis, devices range from task-specific reaching tools [9] to multi-degree-of-freedom systems enabling dexterous manipulation [10].

Despite this diversity, relatively few assistive wearable devices achieve sustained real-world adoption. A primary cause of device abandonment is the lack of personalization. Because assistive devices function as intimate physical interfaces between humans and robots, both functional performance and user comfort are critical [11]. However, individuals with neurological injuries exhibit substantial inter-subject variability, including differences in residual motor function, range of motion, joint contracture, and limb dimensions. Thus, regardless of whether a device is passive or active, task-specific or generalized, effective deployment ultimately depends on tailoring the design to the individual user.

Personalization of assistive devices is therefore essential but challenging due to the high-dimensional, coupled human-device design space. Device parameters (e.g., finger size, number of fingers, phalange configuration, alignment, and actuation layout) interact combinatorially with user-specific functional characteristics. Manual design exploration through iterative prototyping is costly, time-consuming, and difficult to scale.

Computational design methods offer a better path. Recent work has shown that parameterizing device geometry and optimizing over a user-specific objective can meaningfully improve fit [11]. The missing piece is an optimization method that handles the mixed discrete–continuous, geometrically structured design space without requiring a differentiable objective. Bayesian optimization degrades on non-smooth success/failure objectives [12]. Reinforcement learning has no role when the "policy" is the user's own biology. Evolutionary methods require large evaluation budgets that are prohibitive when each evaluation runs a physics simulation. Each of these alternatives also treats performance as a scalar, discarding the per-task failure structure that reveals whether a design fails to contact the object or contacts but cannot lift

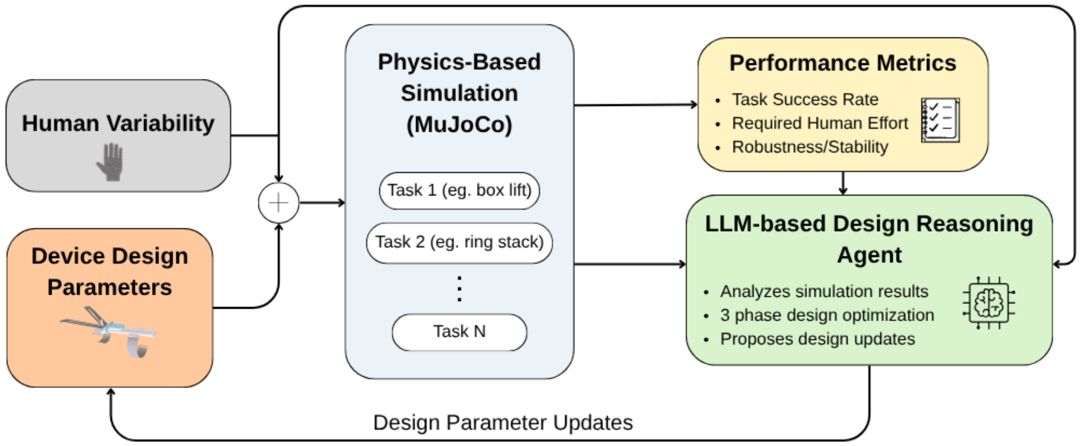

Fig. 1: Overview of the proposed LLM-driven personalized assistive device design framework. Human variability profiles, including hand dimensions, wrist range of motion, and muscle strength, define user-specific constraints for the assistive wearable. Parameterized device designs are evaluated through physics-based MuJoCo simulations across representative tasks, producing abstracted performance metrics such as task success, required human effort, and robustness. A LLM reasons over these simulation outcomes and human constraints to propose design updates, forming a closed-loop, simulation-in-the-loop personalization process.

it — two failures with opposite geometric remedies.

Recent advances in large language models (LLMs) offer a promising alternative for navigating such complex design spaces [13]–[15]. Conventional optimization techniques primarily address numerical parameter tuning, whereas personalization of assistive devices requires structured reasoning over mixed discrete–continuous design variables and coupled human-device interactions. Leveraging the semantic reasoning capabilities and prior knowledge embedded in LLMs, we propose an LLM-driven human-aware design framework that integrates structured parameter generation with physics-based simulation. Through iterative simulation-based evaluation across multiple manipulation tasks and varying human functional conditions, the proposed framework enables scalable personalization of upper-limb assistive wearable devices.

A. Overview

Section II surveys related work on assistive devices, human variability, and AI-driven design. Section III describes the device architecture and 15-parameter design space for C5–C7 SCI users. Section IV details the three-phase LLM pipeline and evaluation metrics. Section V reports outcomes across initialization conditions and user profiles, and Section VI discusses implications and limitations.

The main contributions of this work are as follows:

1) Formulation of assistive device personalization as coupled human–device design space exploration, amenable to LLM-guided search with structured simulation feedback.
2) An LLM-driven human-aware design framework that integrates reasoning-based parameter generation with physics-based simulation evaluation.
3) Demonstration of the effectiveness of the proposed framework across multiple manipulation tasks under varying human functional conditions.

II. Design and Personalization Challenges Across Assistive Devices

A. Personalized Assistive Wearable Devices

There is a wide range of upper-limb assistive devices for individuals with motor impairments. These include passive orthoses, active exoskeletons and supernumerary robotic limbs [4]- [8]. Active systems such as soft robotic gloves directly actuate finger motion to assist grasping, enabling more forceful manipulation but at the cost of added complexity and weight [6]. Supernumerary approaches attach additional robotic fingers to the dorsum of the hand, expanding the user's functional grasping workspace without requiring intact finger function [7], [8].

Despite this variety, a recurring challenge across all device categories is the lack of personalization. Most designs are validated on a narrow set of tasks or user profiles, limiting real-world adoption. Horn [9] developed a universal hand assist tool for patients with reduced upper extremity mobility, highlighting the difficulty of designing a single device that performs reliably across diverse manipulation needs. More dexterous solutions, such as fully wearable actuated soft exoskeletons, have demonstrated broader task coverage [10], but their mechanical complexity makes individualized fitting difficult to achieve without significant engineering effort for each user.

Computational design methods offer a path toward scalable personalization. Abadian et al. [11] proposed a framework for the computational design of personalized wearable robotic limbs, demonstrating that automated optimization can accommodate inter-user variability in limb geometry. Our work builds on this direction by extending computational personalization to functional performance across multiple manipulation tasks, using physics-based simulation and LLM-driven reasoning to navigate the coupled human-device design space.

B. Human Variability in Assistive Robotics

Individuals with neurological injuries such as cervical SCI exhibit substantial inter-subject variability across multiple physical dimensions, including hand size, wrist range of motion, residual muscle strength, and functional joint deviations such as ulnar deviation. This variability directly affects device fit, actuation requirements, and task performance, making one-size-fits-all designs inadequate for clinical deployment [2], [3].

Wrist range of motion and extension strength are particularly critical for passive assistive devices that rely on tenodesis grasp. Individuals at SCI levels C5–C7 retain partial wrist extension but differ considerably in the degree of preserved function, which affects both the force available to actuate a passive device and the angular range over which it can operate [4], [5]. Hand dimensions, particularly the distance from the wrist to the metacarpophalangeal joint, determine the physical fit of any dorsal attachment, while functional deviations such as ulnar deviation alter the effective approach angle during grasping.

These sources of variability are rarely addressed systematically in the device design process. Most prior work fixes device geometry to a representative user or adjusts it manually through clinical fitting, which is time-consuming and difficult to scale across a heterogeneous patient population. The proposed framework addresses this gap by explicitly encoding four variability parameters: hand length, wrist extension, wrist strength, and ulnar deviation as hard constraints within the simulation environment, ensuring that every evaluated design reflects what is physically achievable for the specific user.

C. AI-driven Robot Design

Automated approaches to robot design can be broadly categorized into reinforcement learning-based morphology optimization, Bayesian and evolutionary methods, and more recently, LLM-based design frameworks.

RL enables scalable morphology optimization [16], [17] but requires a trainable policy. Here, actuation is the user's own wrist extension — a biological constant — so RL is inapplicable; the design challenge is purely geometric.

Bayesian optimization (BO) is sample-efficient for smooth continuous objectives [12], but degrades on the non-smooth, mixed discrete–continuous space here and cannot perform the geometry-grounded reasoning needed to diagnose specific failure modes and propose targeted corrections.

Evolutionary methods face two issues here: per-candidate evaluation across eight physics simulations is prohibitively expensive, and their co-optimization power comes from jointly evolving morphology with adaptive behavior [18]. Our device has no evolvable behavior, so they incur full evaluation cost with none of the benefit that justifies it.

More recently, LLMs have been applied to design tasks requiring structured reasoning over geometric parameters. Du et al. [13] trained LLMs for computer-aided design with iterative self-improvement, while Li et al. [14] demonstrated parametric 3D model generation from natural language prompts. Qiao et al. [15] applied language-driven morphological design specifically to robotic hands, showing that LLMs can reason over shape parameters to satisfy functional objectives. Our work extends this line of research to the personalized assistive device domain, where the design space is coupled with user-specific physical constraints and must be optimized against functional performance requirements across multiple manipulation tasks.

III. Assistive Device Design and Human Variability Modeling

A. Target Population and Task Definition

C5–C7 SCI patients retain partial wrist extension but lose finger flexion. The dorsal grasper exploits this: as the wrist extends, it presses the hand against fixed finger structures above the dorsum, pinching objects in between. The device adds no actuation — it works entirely off the user's residual wrist motion.

We define eight manipulation tasks drawn from activities of daily living, spanning a range of object geometries, masses, and manipulation requirements. Tasks are summarized in Table I. The two binding constraints across all successful designs are `ring_stack` and `sphere_lift`: ring stacking demands precise lateral positioning for placement within 25 mm of a rod, and sphere lifting drives peak force requirements close to the user's ceiling.

TABLE I: Eight Evaluation Tasks

Task	Object	Key Requirement
Cylinder Lift	Cylinder	Lift and hold above threshold
Small Cylinder Lift	Small cyl.	Lift and hold (smaller contact area)
Large Cylinder Lift	Large cyl.	Wide finger spread required
Cylinder Lift+Rotate	Cylinder	Lift, rotate 90°, hold
Cylinder Yaw 360°	Cylinder	Sustained grasp through full 360° yaw
Box Lift	Box	Lift and hold
Sphere Lift	Sphere	Lift and hold (high force demand)
Ring Stack	Ring	Lift, translate to rod, release

B. Human Variability Parameters

Each user profile encodes four physical parameters as simulation constraints: (1) wrist pitch ROM — maximum extension angle in radians, setting the kinematic envelope; (2) maximum wrist torque — a hard actuator ceiling in simulation, the most critical constraint for task success; (3) wrist plate length — forearm-to-MCP distance, setting device mounting geometry; and (4) functional deviation (wrist yaw) — resting lateral offset of the wrist axis.

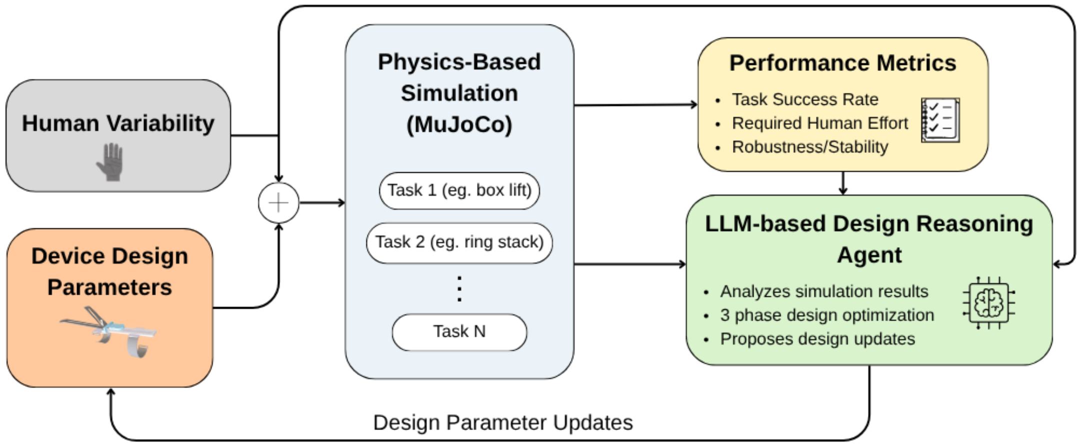

Fig. 2: Overview of the proposed LLM-driven personalized assistive device design framework. Human variability profiles, including hand dimensions, wrist range of motion, and muscle strength, define user-specific constraints for the assistive wearable. Parameterized device designs are evaluated through physics-based MuJoCo simulations across representative tasks, producing abstracted performance metrics such as task success, required human effort, and robustness. A LLM reasons over these simulation outcomes and human constraints to propose design updates, forming a closed-loop, simulation-in-the-loop personalization process.

Ten user profiles (H01–H10) were defined and collected from real clinical measurements listed in Table II. Experiments were run on H01–H04; H05–H10 are defined for future work.

TABLE II: Human Variability Parameters (H01–H10)

Profile	Plate [cm]	UEMS (Force)	Wrist Ext. [°]	Ulnar Dev. [°]
H01	10.0	5 (2.92 Nm)	69	—
H02	10.0	5 (2.92 Nm)	40	—
H03	9.5	3 (1.92 Nm)	65	—
H04	9.8	2 (1.42 Nm)	63	—
H05	9.0	5 (2.92 Nm)	62	24
H06	9.0	5 (2.92 Nm)	72	—
H07	10.0	5 (2.92 Nm)	86	—
H08	10.0	1 (0.92 Nm)	34	—
H09	10.0	5 (2.92 Nm)	70	—
H10	10.0	5 (2.92 Nm)	80	—

C. Targeted Assistive Device: Passive Dorsal Grasper

The passive dorsal grasper consists of a wrist-mounted plate secured with ventral straps and two rigid finger chains projecting above the hand in a V-configuration. No motors, cables, or sensors. When the wrist extends, the hand surface rises and closes against the fingers; objects caught in this pinch can be lifted and manipulated using wrist motion alone.

This device class was selected because its mechanical simplicity makes it directly parameterizable: the geometry is fully described by a small set of continuous and discrete variables with clear physical meaning. The dorsal mounting platform was validated for C5–C7 users in prior work [8]. An image of the device and its labeled geometry is shown in Fig. 3.

D. Assistive Device Design Parameterization

The full geometry is described by 15 parameters:

Topology: $\texttt{num\_phalanges} \in \{1, 2, 3, 4, 5\}$ — segments per finger chain. This is the only discrete parameter.

Per-segment geometry (for $k = 1$ to $\texttt{num\_phalanges}$): θ_k is the pitch angle of segment k relative to segment $k-1$ (or the plate for $k = 1$); l_k is the segment length in meters, with l_1 being the proximal segment and always the longest.

Global geometry: phalange width w; finger spread angle φ_{spread} (half-angle of the V-opening); finger start position $p_{\text{start}} \in [0, 1]$ (normalized attachment position along the plate, 0 = center, 1 = tip); finger yaw angle φ_{yaw} (lateral rotation to accommodate wrist deviation).

For n phalanges, the active parameter count is $2n+4$. Full bounds are listed in Table III.

TABLE III: Design Parameter Bounds and Defaults

Parameter	Min	Max	Default	Unit
num_phalanges	1	5	2	count
θ_1	0.05	3.14	0.51	rad
θ_2–θ_5	0.10	1.20	0.6	rad
l_1	0.05	0.25	0.119	m
l_2–l_5	0.015	0.10	0.019	m
w	0.006	0.050	0.012	m
φ_{spread}	0.00	0.50	0.17	rad
p_{start}	0.0	1.0	0.50	—
φ_{yaw}	−0.50	0.50	0.00	rad

Parameters for inactive segments ($k > \texttt{num\_phalanges}$) have no physical effect. Bounds were verified to contain known good solutions via preliminary feasibility sweeps; minimum dimensions reflect 3D printing constraints.

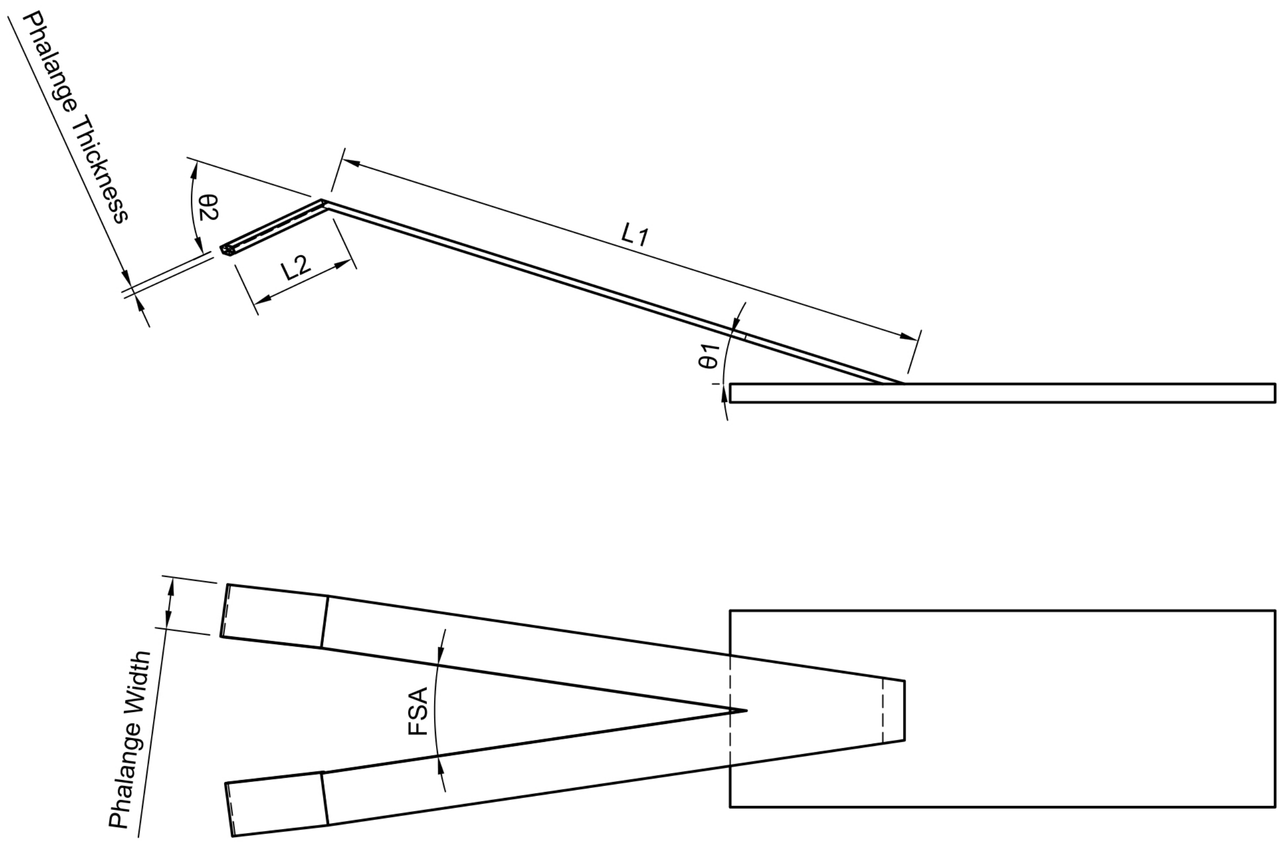

Fig. 3: Passive dorsal grasper design with labeled geometric parameters. Two symmetric finger chains extend from the wrist plate, forming a V-opening above the hand. Each chain is defined by segment angles θ_k and lengths l_k. The finger spread angle φ_{spread} controls the V-opening width.

IV. LLM-Driven Personalized Design Framework

A. Overview of the Closed-Loop Design Process

The pipeline runs three phases sequentially. At each iteration, design parameters are proposed, a parameterized MuJoCo simulation model is generated, performance is evaluated across the relevant task set, and the LLM receives a structured summary to propose the next candidate.

Phase 1 — Per-task optimization. For each of the 8 tasks independently, the LLM iteratively proposes designs and receives simulation feedback. The loop stops when 3 successful designs are found or the iteration cap is reached (10–30 iterations depending on experimental condition). Each task builds its own candidate pool.

Phase 2 — Cross-validation. All successful Phase 1 designs are evaluated on all 8 tasks. If any passes all 8 (Case A), it becomes the final output and Phase 3 is skipped. Otherwise, the highest-scoring design seeds Phase 3. Two of our 12 runs exited here — their Phase 1 designs generalized to all tasks without explicit multi-task optimization.

Phase 3 — Multi-task optimization. Starting from the Phase 2 seed, the LLM proposes designs evaluated on all 8 tasks simultaneously. The loop runs until 3 consecutive designs achieve universal success, or the budget (50–100 iterations) is exhausted.

The full algorithm is given in Algorithm. 1.

Why LLM-guided search rather than classical optimization. Collapsing 8 per-task outcomes into a scalar objective destroys the semantic content that drives targeted fixes: `no_contact` and `weak_contact` on the same task demand geometrically opposite responses, but a surrogate fitted to an aggregate score cannot recover this distinction [12], [19]. The mixed integer-continuous space (`num_phalanges` $\in \{1, \dots, 5\}$) further degrades surrogate quality above a few continuous dimensions. Reinforcement learning requires a trainable policy, but the actuating motion is the user's own wrist extension — a biological constant, not a learnable function [16]. LLMs instead carry geometric priors from pre-training on engineering literature, enabling sensible proposals without the warm-up data any surrogate-based method requires [13], [20]. The empirical case is decisive: 3,000 random configurations produced zero universal designs; the LLM pipeline converges in 83% of runs within 60 iterations.

B. Simulation-based Evaluation Metrics

1) Task Success: Each task is governed by a finite state machine: REACH $\rightarrow$ CLOSE_GRIPPER $\rightarrow$ LIFT $\rightarrow$ HOLD, with task-specific extensions (ROTATE, MOVE_TO_ROD, RELEASE). A task passes only if every phase meets its success

Algorithm 1 Three-Phase LLM-Driven Design Pipeline

Require: User profile $\mathcal{U}$, tasks $\mathcal{T}$, budgets B_1, B_3
Ensure: Universal design d^* passing all $\mathcal{T}$ within $\mathcal{U}$'s force limit
Phase 1: for each $t_i \in \mathcal{T}$ **do**
  $d_0 \leftarrow$ init; $H \leftarrow \emptyset$; $n_{\text{succ}} \leftarrow 0$
  for $k = 1$ **to** B_1 (or until $n_{\text{succ}} = 3$) **do**
    evaluate d_k on $t_i \rightarrow$ metrics m_k
    $H \leftarrow H \cup \{(d_k, m_k)\}$
    $d_{k+1} \leftarrow \text{LLM}(\mathcal{U}, t_i, H)$
  collect S_i (successful designs for t_i)
Phase 2: for each $d \in \bigcup_i S_i$ **do**
  score$(d) \leftarrow \#\{$tasks passed on all $\mathcal{T}\}$
  if score$(d) = 8$: **return** d ▷ Case A
  $d_{\text{seed}} \leftarrow \arg\max_d$ score(d)
Phase 3: $d \leftarrow d_{\text{seed}}$; $H \leftarrow \emptyset$; $n_{\text{consec}} \leftarrow 0$
  for $k = 1$ **to** B_3 **do**
    evaluate d on all $\mathcal{T} \rightarrow \{m_i\}$
    if all m_i pass: n_{consec} += 1
      if $n_{\text{consec}} \geq 3$: **return** d
    else: $n_{\text{consec}} \leftarrow 0$
    $H \leftarrow H \cup \{(d, \{m_i\})\}$
    $d \leftarrow \text{LLM}(\mathcal{U}, \mathcal{T}, H)$

criterion — lift tasks require the object to reach and maintain a minimum height; ring stack additionally requires placement within 25 mm of the rod center. Failure terminates the task immediately. A rule-based classifier assigns one of seven failure codes: These codes are the primary diagnostic input

TABLE IV: Task Failure Mode Taxonomy

Code	**Meaning**
`no_contact`	Device never contacts the object
`unstable_contact`	Contact made but grip immediately lost
`weak_contact`	Force insufficient at user's maximum torque
`lift_failed`	Grasped but does not reach lift threshold
`dropped_hold`	Reaches height, then drops during hold
`reach_blocked`	Arm cannot reach due to joint limit
`timeout`	Phases do not complete within time budget

to the LLM, distinguishing — for example — whether ring stack failure is a reach problem or a grasp geometry problem.

2) Required Human Effort: Each design evaluation includes a force sweep: wrist pitch torque varies from a task-specific minimum up to the user's maximum, and the lowest torque at which the task succeeds is recorded. This directly measures how far below the user's physical ceiling the design operates.

3) Robustness: Robustness is evaluated by adding small random disturbances to the control signals during simulations.

Actuator control signals receive zero-mean Gaussian noise at every physics timestep:

$$u_{\text{noisy}} = \text{clip}(u + \epsilon,\ u_{\min},\ u_{\max}), \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \tag{1}$$

where σ is in Nm (absolute). A design's *noise tolerance* is the highest σ at which it passes all 8 tasks. We test $\sigma \in \{0, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0\}$ Nm. This models hand tremor and involuntary wrist motion relevant to the SCI population.

C. LLM as a Design Reasoning Agent

1) Inputs: At each iteration, our LLM receives four structured inputs:

- **User profile** (forearm radius, plate length, wrist pitch ROM, max force, functional deviation)
- **Object specification** (shape, dimensions, mass, friction)
- **Full design history** (every prior candidate with its outcome, minimum grip force, peak contact force, contact height fraction 0–1, and failure code)
- **Parameter bounds** for all 15 variables

2) Output: The LLM returns a JSON object with three fields: `parameters` (the proposed design), `reasoning` (a geometry-based explanation referencing specific failure types and contact patterns), and `expected_improvements` (which metrics the LLM predicts will improve). This structured output enables logging not just what was proposed but why, which is essential for analyzing failure modes in the search itself.

3) Reasoning Strategy: The prompt instructs LLM to act as a geometry-aware design agent that makes the smallest change addressing the observed failure without disturbing parameters that are already working. In practice the LLM distinguishes failure types meaningfully. A `lift_failed` failure with high contact force on `cylinder_lift_large` prompts wider finger spread: *"the object is being contacted but slides out; fingers are too narrow for this radius."* A `no_contact` on `ring_stack` prompts reduction in θ_1 to lower the approach angle and a shift in p_{start} toward the plate tip. These are geometrically correct diagnoses. Section V discusses cases where this reasoning breaks down.

We run GPT-4.1 [21] for cheap computational experiments at temperature 0.3.

D. Practical Constraints and Safety Bounds

All 15 parameters are constrained within the bounds in Table III, enforced both in the LLM prompt (via explicit JSON schema validation) and in post-processing before simulation. Phalange thickness and contact friction coefficients are fixed across all experiments.

MuJoCo was selected as the physics modeling simulation environment. Physical plausibility of simulation results against hardware measurements has not yet been validated — that is planned as the next step and is discussed in Section VII.

V. Performance of LLM-Driven Design Pipeline

A. Performance Across Human Variability

Multi-user experiments were run for H01–H04 with a standardized configuration: default parameter initialization, 30-iteration Phase 1 cap, 60-iteration Phase 3 budget. Results are in Table V.

TABLE V: Multi-User Experiment Outcomes

User	UEMS	Force	ROM	Phase 3 Convergence
H01	5	2.92 Nm	69°	iter 45
H02	5	2.92 Nm	40°	(no convergence)
H03	3	1.92 Nm	65°	iter 47
H04	2	1.42 Nm	63°	iter 2

Three of four users converged. H04 converged fastest despite the lowest force cap, which suggests wrist ROM is a more important convergence driver than force limit in this range. H02 is the outlier: same force limit as H01 but only 40° of extension versus 69°. The restricted ROM appears to shrink the feasible design space, requiring more iterations or a different search strategy.

Across all 12 experimental runs (H01 robustness and consistency studies combined with multi-user runs), Fig. 5 shows that **10/12 converged achieving 83.3% convergence rate**.

Phase 1 per-task convergence shows a consistent difficulty ordering. Box lift and small cylinder lift reach 100% convergence across all runs. Cylinder yaw 360° is hardest at 58%, consistent with its demanding sustained-contact geometry. Five of eight tasks exceed 75%. The results are shown in Fig. 4.

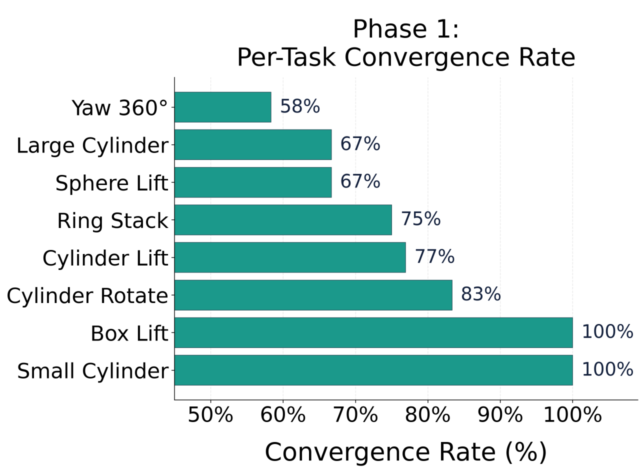

Fig. 4: Phase 1 per-task convergence rates across all experimental runs. Box lift and small cylinder lift achieve 100%; cylinder yaw 360° converges in 58% of runs. Five of eight tasks exceed 75%.

B. Effect of Iterative LLM-Driven Design Updates

1) Robustness to Initialization: Five runs tested whether the pipeline succeeds from diverse starting points (one default, four random initializations). Results are in Table VI.

TABLE VI: Robustness Study (H01, 5 Runs Varying Initialization)

Run	Init	P1 Tasks	P2 Best	P3 Result
1	Default	8/8	7/8	✓ iter 45
2	Random	8/8	8/8 (Case A)	— skipped
3	Random	6/8	7/8	✓ iter 2
4	Random	3/8	7/8	✓ iter 30
5	Random	2/8	8/8 (Case A)	— skipped

5/5 converged (100%). The most striking result is Run 4: starting with only 3/8 tasks solvable in Phase 1, the pipeline still found a universal design at Phase 3 iteration 30. Phase 2 cross-validation designs from those three tasks happened to generalize to four more, producing a 7/8 Phase 2 seed. This is the strongest evidence that LLM reasoning compensates for poor initialization.

We also compared LLM-guided search against random parameter sampling (uniform sampling within bounds, no LLM). A 3,000-configuration sweep produced **zero universal designs** (0%); the best random configurations reached 7/8 tasks only twice. The LLM pipeline converged in 83% of runs within 60 iterations as shown in Fig. 5. The effect of iteration budget on LLM non-determinism across fixed-initialization consistency runs is detailed in Table VII (Appendix A). A representative fast-convergence Phase 3 trajectory is given in Table VIII (Appendix B).

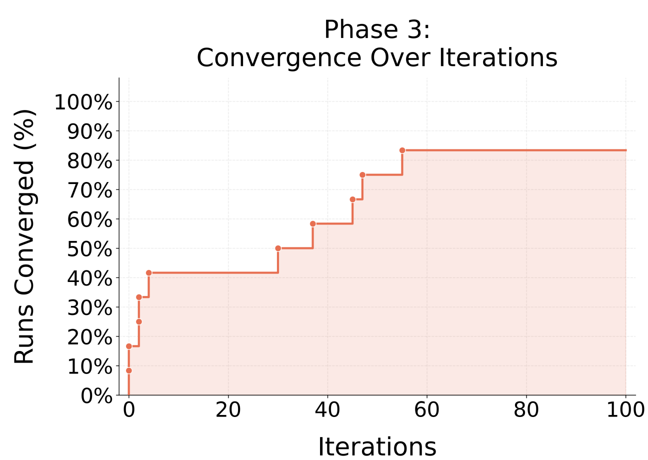

Fig. 5: Runs converged percentage across iterations in phase 3. Phase 3 was tested each design iteration on all 8 tasks until 50 total iterations or convergence. 17% of trials were already converged after Phase 2. By 50 iterations, the number of converged runs equaled approximately 83%.

2) Universal Design Convergence Zone: Across all 8 confirmed winning designs, parameter values cluster consistently regardless of initialization. This result is shown in Table IX. A simulation render of the fastest-converging instance—`exp_consistency_v2_r3` Phase 3 iteration 4—is shown in Fig. 6; it is representative of all confirmed universal designs.

3) Force Requirements: Across all successful designs, 7 of 8 tasks require a mean minimum grip force of $\leq$1.8 Nm, well within the 2.92 Nm H01 ceiling. The binding constraints

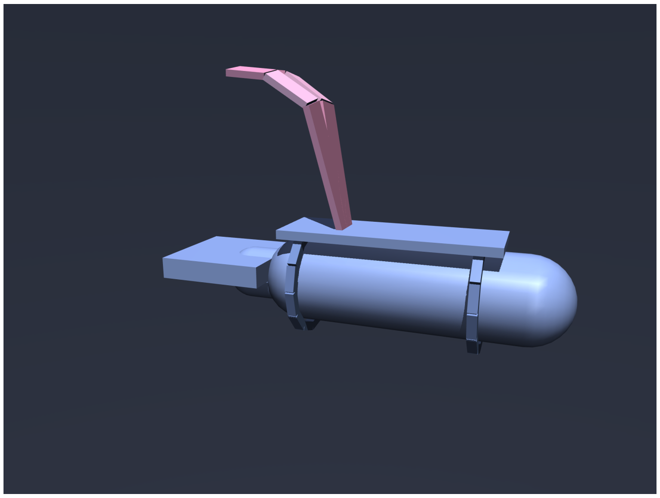

Fig. 6: MuJoCo simulation render of the fastest-converging universal design (`exp_consistency_v2_r3`, Phase 3 iter. 4). Parameters: n_p=3 phalanges, θ_1=1.35 rad, φ_{spread}=0.15 rad, l_1=60 mm. The V-shaped two-finger chain (pink) mounts dorsally on the wrist-forearm module. When the wrist extends, the palmar surface rises into contact with the fingers, enabling passive grasping without motors or cables.

are sphere lift and ring stack, which consistently require 2.3–2.92 Nm. The best Phase 3 design achieves both at 2.3 Nm, a 0.62 Nm margin below H01's ceiling. Full per-task force distributions across all winning designs are shown in Fig. 7.

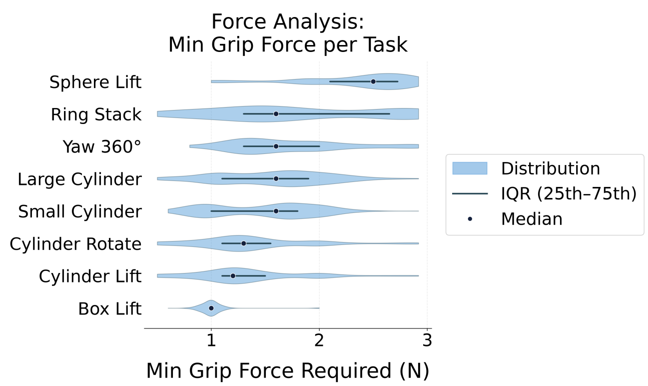

Fig. 7: Distribution of minimum required grip force per task across all successful H01 designs. Sphere lift and ring stack are the binding constraints; 7 of 8 tasks are achievable at a mean ≤1.8 Nm.

C. Noise Robustness Analysis

We tested all converged designs under actuator noise per eq. 1 at $\sigma \in \{0, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0\}$ Nm (9 experiments, 3 design variants each).

Converged designs are generally noise-tolerant. Most designs pass 7 or more tasks at $\sigma = 0.5$ Nm; several remain fully functional through $\sigma = 1.0$ Nm (Table X). The most noise-robust design, one whose parameters were set conservatively, leaving 0.3 Nm of headroom on sphere lift and ring stack relative to H01's force cap, passes all 8 tasks through $\sigma = 2.0$ Nm. Results are shown in Fig. 8.

$\sigma = 5.0$ **Nm is the universal failure cliff.** Every design collapses to 0/8 tasks at $\sigma = 5.0$. The transition is sharp.

ring_stack fails first. In every experiment ring_stack has the lowest or tied-lowest noise tolerance. The policy requires placing the ring within 25 mm of the rod, any actuator perturbation that shifts the finger trajectory causes the ring to slip or miss. First failure tasks are show in Table X

cylinder_yaw_360 is the second most fragile. Sustained contact over a full 360° rotation means noise accumulates. Users with tighter force ceilings (H03, H04) show yaw_360 failures at $\sigma = 0.5$ even on designs that otherwise survive to $\sigma = 1.0$.

Force margin determines noise tolerance. Designs where sphere lift and ring stack requirements sit well below the user's force cap absorb perturbations without failing. The most robust design was originally optimized for a low-force task (small cylinder lift) and its conservative geometry leaves 0.3 Nm of margin on the two hardest tasks. This margin survives as 2.0 Nm noise headroom. Designs tuned to pass all 8 tasks at the exact force ceiling leave no such margin.

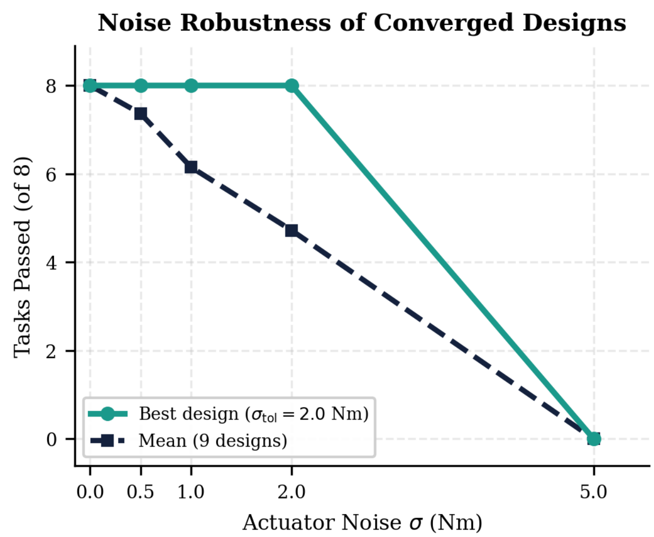

Fig. 8: Noise robustness of all converged designs. Faint gray lines: individual experiments. Teal solid: most robust design ($\sigma_{\text{tol}} = 2.0$ Nm). Navy dashed: mean across all 9 designs. All designs converge to 0 tasks at $\sigma = 5.0$ Nm.

VI. Discussion

LLM reasoning is effective but anchors. The 83% convergence rate, against zero successes in 3,000 random samples, confirms the LLM's geometry-aware reasoning provides genuine value. All winning designs converge to a consistent region of parameter space ($\theta_1 \in [1.25, 1.48]$ rad, $\varphi_{\text{spread}} \in [0.09, 0.19]$ rad) despite diverse initializations; a representative design from this convergence zone is rendered in Fig. 6. The LLM has internalized geometric priors about grasp stability that guide search toward the correct region. But the primary failure mode across every non-converging run is parameter anchoring: the LLM fixes on $\varphi_{\text{spread}} = 0.09$ rad and does not escape even with 100 iterations. The failure code it sees (`lift_failed`, high contact force) is ambiguous, it could indicate geometric instability (the correct diagnosis: need wider spread) or insufficient user force (incorrect: the force sweep already tests up to the user's cap). The LLM apparently gravitates toward the second interpretation, varying l_3, θ_2, and p_{start} rather than the critical parameter. This is a reasoning error, not a budget limitation; doubling the Phase 3 budget to 100 iterations did nothing.

Phase 2 generalization is a recovery mechanism. In one high-budget run, sphere lift was never solved across 30 Phase 1 iterations. Yet the Phase 2 best design (optimized for large cylinder lift) passed sphere lift in cross-evaluation. The geometric properties that enable large-cylinder grasping, sufficient spread angle and phalange curvature, also provide adequate sphere contact. This indicates that a missing Phase 1 task is not always fatal; cross-task generalization in Phase 2 can rescue it.

Noise robustness validates real-world applicability. Most converged designs remain functional through σ = 0.5 Nm, consistent with actuator variability and the hand tremor levels typical of SCI users. The most noise-tolerant design passes all 8 tasks at $\sigma = 2.0$ Nm because its force requirements on sphere lift and ring stack sit 0.3 Nm below H01's cap, leaving a margin that persists as noise headroom. The standard success criterion optimizes for minimum required force but not for robustness. Adding a noise-tolerance requirement, require passing at $\sigma = 0.5$ Nm rather than $\sigma = 0$, would likely produce designs that are both force-efficient and practically deployable, at some cost in convergence speed.

Wrist ROM matters more than force limit. H04 has the lowest force cap (1.42 Nm) yet converged in 2 iterations. H02 has H01's force cap but only 40° of ROM and failed to converge. Reduced ROM restricts the kinematic envelope, making some finger approach angles geometrically impossible. The feasible design space contracts. With only four users, this observation is suggestive rather than conclusive, but it has practical implications: users with severely restricted ROM (like H02) may not have sufficient kinematic range for any passive device geometry, and an active device extension would be required.

Limitations. All results are simulation-only; the MuJoCo contact model uses fixed friction coefficients that will not match real surfaces. Task policies approach objects from fixed positions, not from the naturalistic trajectories humans use. The cohort of four tested users is too small for population-level conclusions. A ring_stack false positive identified in one default-initialization run means its ring_stack results are unreliable throughout Phase 1 (the failure has since been traced to a policy bug in which lost ring tracking is misreported as task success).

VII. Conclusion

A passive dorsal grasper can be parameterized in 15 variables and optimized for a specific user through LLM-guided physics simulation. Across 12 experiments and four users, the three-phase pipeline finds universal designs, passing all 8 tasks within each user's force ceiling, 83% of the time. The fastest convergence is 2 iterations. Random parameter search produced zero universal designs in 250 attempts.

The dominant failure mode is LLM parameter anchoring: the model locks onto a suboptimal φ_{spread} and does not escape even with 100 additional iterations. This is fixable with targeted prompt engineering. Noise robustness testing shows that designs with conservative force margins tolerate actuator noise significantly better than designs that operate near the user's force ceiling — mean tolerance across all converged designs is 0.5 Nm, with one outlier reaching 2.0 Nm. For physical fabrication, that most noise-tolerant design is the strongest candidate.

A. Future Work

Sim-to-Real (Main) As the results in the project are completely simulation-based, our next step would be to manufacture the optimal designs and test whether the wrist force required and the contact force measured are similar to our simulated values. Tests should also be conducted to see if patients perceived ease of use are improved with the personalized designs.

Prompt injection for critical parameters. For `ring_stack lift_failed` failures, embed explicit geometric guidance that $\varphi_{\text{spread}} < 0.10$ rad is rarely sufficient. This directly targets the primary failure mode at zero architectural cost.

Noise-aware success criterion. Require Phase 3 designs to pass all tasks at $\sigma = 0.5$ Nm rather than $\sigma = 0$. This will slow convergence but produce designs with practical noise headroom for real-world use.

Physical fabrication and validation. The most noise-tolerant converged design (sphere lift and ring stack at 2.6/2.7 Nm; noise tolerance $\sigma = 2.0$ Nm) is the strongest fabrication candidate. Physical force measurements will establish whether MuJoCo predictions hold and close the simulation-to-reality gap.

Realistic task trajectories. Current policies use straight-line approaches. Recording actual human reaching motions and using them as simulation trajectories would make evaluations more representative of daily-use conditions.

Extended user cohort. H05–H10 profiles are defined; completing those runs would clarify the relationship between wrist ROM and convergence rate and enable population-level conclusions.

Active device extension. Users with severely restricted ROM (e.g., H02 at 40°, H08 at 34°) may lack sufficient kinematic range for any passive geometry. An actuated version of the dorsal grasper would expand the accessible design space for these users.

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Appendix

A. LLM Non-determinism and Iteration Budget

Two consistency test batches from identical default parameters quantified the effect of search budget, summarized in Table VII. Doubling both budgets improved success from 33% to 67%, but did not eliminate failures. One high-budget run had the strongest possible Phase 1 (8/8 tasks), a 7/8 Phase 2 seed, and the full 100-iteration Phase 3 budget — and still failed. The culprit was φ_{spread}: the LLM proposed 0.09 rad in 64 of 100 Phase 3 iterations, varying everything else while leaving this parameter fixed. Every universal design found requires $\varphi_{\text{spread}} \geq 0.11$ rad. The LLM was locked 0.02 rad below the critical threshold for the entire run. Doubling the budget did not help because the issue was not budget — it was the LLM anchoring on a locally plausible but globally wrong value.

TABLE VII: Consistency Tests — Effect of Iteration Budget (3 runs each, default initialization, H01)

Budget	P1 Cap	P3 Budget	Success
Low	10 iter	50 iter	1/3 (33%)
High	30 iter	100 iter	2/3 (67%)

B. Fast-Convergence Trajectory Example

The two fastest convergences (at iterations 2 and 4, respectively) occurred when the LLM immediately targeted both θ_1 and φ_{spread} together. Table VIII traces one such run from a poor 6/8 starting point to a universal design in four Phase 3 iterations.

TABLE VIII: Phase 3 Fast Convergence Example (4 iterations to universal design)

Iter	φ_{spread}	θ_1	ring_stack	All tasks
1	0.12	1.65	FAIL	6/8
3	0.13	1.35	FAIL	7/8
4	**0.15**	**1.35**	**PASS**	**8/8 ✓**

C. Universal Design Convergence Zone

Table IX shows the parameter bounds shared by all eight confirmed universal designs, regardless of initialization or user profile. The range for φ_{spread} has a hard lower bound of 0.11 rad imposed by the ring_stack task; every failed run anchored below this threshold.

TABLE IX: Universal Design Convergence Zone

Parameter	Working Range	Key Observation
θ_1	1.25–1.48 rad	No winner outside this range
φ_{spread}	0.09–0.19 rad	ring_stack requires ≥ 0.11
p_{start}	0.37–0.62	Wider than expected
`num_phalanges`	2 or 3	3 gives more force headroom
l_1	0.055–0.072 m	Consistent across all
φ_{yaw}	-0.04 to $+0.08$ rad	Near-zero in all cases

D. Noise Tolerance by Design

Table X reports the noise tolerance threshold and first-failure task for every converged design across all participants and studies. Conv. denotes the convergence point: cross-val. indicates the design was selected during Phase 2 cross-validation; iter N denotes the Phase 3 iteration at which convergence was confirmed.

TABLE X: Noise tolerance of all converged designs.

User	Init	Conv.	Tol. σ (Nm)	First Failure
H01	Random	cross-val.	**2.0**	none at $\sigma \leq 2.0$
H01	Random	cross-val.	1.0	ring_stack
H01	Random	iter 2	1.0	ring_stack
H01	Default	iter 55	0.5	ring_stack, yaw_360
H01	Default	iter 37	0.0	ring_stack, sphere_lift
H01	Default	iter 4	0.0	ring_stack, sphere_lift
H01	Random	iter 30	0.0	ring_stack, yaw_360
H03	Default	iter 47	0.0	ring_stack
H04	Default	iter 2	0.0	ring_stack, yaw_360